Based on (http://sleipnir.cs.csubak.edu/~erangel/cmps2240-f17/doc/WGC_Chapter_3.pdf)

Multiplying Binary Values

Multiplication of binary numbers is also very easy. It's just like decimal multiplication involving only zeros and ones (which is trivial). Here are the rules you need to know for binary multiplication:

- 0 × 0 = 0
- 0 × 1 = 0
- 1 × 0 = 0
- 1 × 1 = 1

Using these four rules, multiplication is done the same way you'd do decimal multiplication (in fact, if you just follow the rules for decimal multiplication on your binary values you'll actually get the correct results, because the rules for decimal multiplication involving the zero and one digits are identical). Here are some examples of binary multiplication:

```
      1010
   ×  0101
   -------

Step 1: Multiply the LO bit of the multiplier times the multiplicand.

      1010
   ×  0101
   -------
      1010    (1 × 1010)

Step 2: Multiply bit one of the multiplier times the multiplicand.

      1010
   ×  0101
   -------
      1010    (1 × 1010)
     0000     (0 × 1010)
   -------
     01010    (partial sum)

Step 3: Multiply bit two of the multiplier times the multiplicand.

      1010
   ×  0101
   -------
    001010    (previous partial sum)
    1010      (1 × 1010)
   -------
    110010    (partial sum)

Step 4: Multiply bit three of the multiplier times the multiplicand.

      1010
   ×  0101
   -------
    110010    (previous partial sum)
   0000       (0 × 1010)
   -------
   0110010    (product)
```

Dividing Binary Values

Like multiplication of binary numbers, binary division is actually easier than decimal division. You use the same (longhand) division algorithm, but binary division is easier because you can trivially determine whether the divisor goes into the dividend during each step of the longhand division algorithm. Figure 3-1 on the next page shows the steps in a decimal division problem.

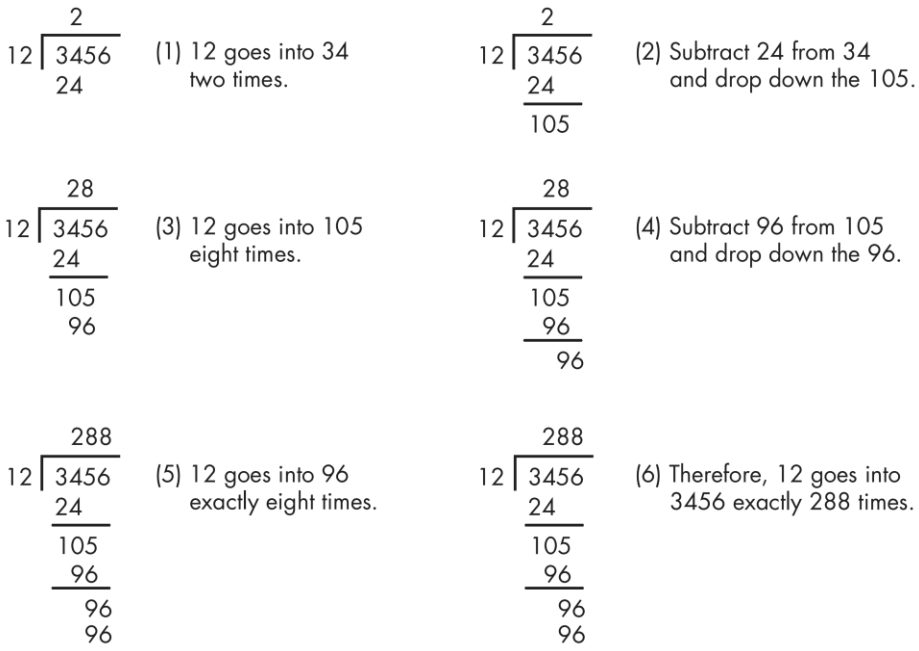
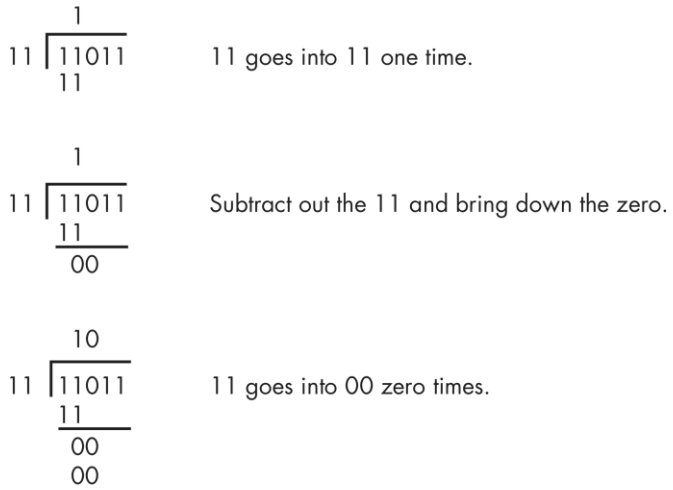


Figure 3-1: Decimal division (3456/12)

This algorithm is actually easier in binary because at each step you do not have to guess how many times 12 goes into the remainder nor do you have to multiply 12 by your guess to obtain the amount to subtract. At each step in the binary algorithm, the divisor goes into the remainder exactly zero or one times. As an example, consider the division of 27 (11011) by three (11) as shown in Figure 3-2.



```
      10
   ------
11 | 11011
     11
     --
     00
     00
     --
      01
```

Subtract out the zero and bring down the one.

```
      100
   ------
11 | 11011
     11
     --
     00
     00
     --
      01
      00
```

11 goes into 01 zero times.

```
      100
   ------
11 | 11011
     11
     --
     00
     00
     --
      01
      00
      --
       11
```

Subtract out the zero and bring down the one.

```
      1001
   ------
11 | 11011
     11
     --
     00
     00
     --
      01
      00
      --
       11
       11
```

11 goes into 11 one time.

```
      1001
   ------
11 | 11011
     11
     --
     00
     00
     --
      01
      00
      --
       11
       11
       --
       00
```

This produces the final result of 1001.

Figure 3-2: Longhand division in binary